Question 1 – Ocen Aaron

An association of secondary schools wants a system to capture students' PLE results.

a) Write a C program that allows **each student to input** their Name, Index Number, Gender, and marks in 4 subjects. Use a loop to allow input for multiple students.

b) Write the **possible output** for one student entry.

c) Explain how the **loop you used in (a)** works.

d) Write a C program that calculates the **average marks** for each student entered.

e) Using an **if-else conditional**, write a C program to assign **grades** for each subject. Use a loop for all subjects.

f) Show the **output of the grade assignment program** for one student.

g) Draw a **flowchart** showing how the if-else conditional works.

h) Write a C program using a **switch construct** that determines the school choice based on the average marks.

i) Write short notes for **at most 5 tokens** you have used in (a)–(h).

j) Rewrite the program in (a) using a **different loop** (e.g., if you used for, now use while).

k) Mention at least **two relational operators** used in the programs above.

Question 2 – Nakirayi Robinah

A library wants to track book borrowing by students.

a) Write a C program to **allow students to enter their ID, name, and the book they want to borrow**. Use a loop to input data for **5 students**. (8 marks)

b) Write a function checkAvailability() that **checks if the book is available** and returns 1 if yes, 0 if no.

c) Use **if-else statements** to display whether the student can borrow the book or not based on the function above.

d) Write the **output** if one student borrows a book that is not available.

e) Rewrite the program using a **switch statement** to categorize students based on their grade level (1 – Primary 7, 2 – S1, 3 – S2, 4 – S3).

Question 3 – Kiconco Samuel

A school wants to analyze the exam results of students.

a) Write a C program to **input marks of 5 students** in **3 subjects** using arrays. (8 marks)

b) Compute the **total and average marks** for each student using loops.

c) Write a **function** highestMarks() that finds the student with the **highest total marks**.

d) Write a **function** lowestMarks() that finds the student with the **lowest total marks**.

e) Display the **output** showing each student's marks, total, average, highest, and lowest marks.

f) Rewrite the program to **also calculate the class average for each subject**.

Question 4 – Kamusiime Bridget

A teacher wants a program to generate multiplication tables.

a) Write a C program using a **function** printTable(int n) that prints the multiplication table of number n.

b) Use a **for loop** in main to print tables for numbers **1 to 10**.

c) Rewrite the program to use a **while loop** instead of for.

d) Explain how the loop works in your program.

e) Show the **possible output** for the table of 5.

Question 5 – Baryanengwe Buharl

A school wants to track student attendance.

a) Write a C program to **input attendance for 10 students** (1 = present, 0 = absent) using an array.

b) Calculate the **total number of students present** using a loop.

c) Write a function attendanceStatus(int n) that prints **"Good" if present >= 8, "Average" if 5–7, "Poor" if <5**.

d) Use **if-else** inside the function for decision making.

e) Display the **output for all 10 students**.

Question 6 – Ilakut Juventine

a) Write a C program to **input marks of 5 students in 4 subjects**.

b) Compute **average marks** for each student using a function.

c) Sort the students by **average marks in descending order** using arrays.

d) Display the **student names and average marks in rank order**.

e) Rewrite the program using a **function to swap student data** during sorting.

Question 7 – Kobugabe Bridget

a) Write a C program to **input fees paid by 10 students**.

b) Determine which students **paid in full**, which paid **partially**, and which **did not pay** using if-else.

c) Count the number of students in each category using loops.

d) Display **output showing each student's payment status**.

Question 8 – Nakakawa Maria

a) Write a C program to **input the number of days each student delayed returning a book** (array of 10 students).

b) Write a function calculateFine(int days) that returns **fine = days * 200** if days > 0, else 0.

c) Display **fine for each student using a loop**.

d) Calculate **total fines collected** using a function.

Question 9 - Kyalisima Angella

a) Write a C program to **input marks for 5 students in 5 subjects**.

b) Write a function computeGrade(int mark) to assign grades:

- 80–100: A
- 60–79: B
- 40–59: C
- <40: F

c) Use a loop to **display all students' marks and grades**.

d) Rewrite the program to **also display the student with the highest average**.

Question 10 – Muhumuza Peter

a) Write a C program to **input votes for 4 candidates** from 20 students.

b) Use arrays to **count votes per candidate**.

c) Use if-else to **declare the winner**.

d) Display **total votes and winner**.

Question 11 – Musika Alvin

A library wants to track books issued to students.

a) Write a C program to allow input of **Student Name, ID, and Number of Books Issued** for 5 students. Use a loop. (8 marks)
b) Display all student details after input.
c) Explain how the loop works in your program.
d) Write a function totalBooks() that calculates the **total number of books issued**.
e) Write a function using **if-else** to flag students who have issued **more than 3 books**.
f) Display the output for one sample student.

Question 12 – Birimawa Patience

A school wants to analyze student attendance.

a) Input attendance (1 = present, 0 = absent) for **10 students** using an array.
b) Write a function to **calculate total students present**.
c) Write a function to **calculate total students absent**.
d) Use a loop to display **each student's attendance status** (Present/Absent).
e) Draw a **flowchart** for the attendance check.

Question 13 – Munguchi Yasir

A school wants to compute exam results.

a) Input marks for **5 students in 4 subjects** using arrays. (8 marks)
b) Write a function computeTotal(int marks[]) to compute total marks for a student.
c) Write a function computeAverage(int total) to compute average marks.
d) Using if-else, assign grades (A, B, C, F).
e) Display a **result table** showing Name, Total, Average, Grade.

Question 14 – Niwabikora Promise

A school wants to track fee payments.

a) Input fee paid by **10 students**.
b) Write a function using **if-else** to determine payment status: Full (>=500,000), Partial (100,000–499,999), None (<100,000).
c) Display the **number of students in each category** using loops.
d) Display the **output for all students**.

Question 15 – Jeero Benon

a) Write a function printTable(int n) to print the multiplication table of a number.
b) Using a loop, print tables for numbers 1 to 5.
c) Rewrite the program using a **while loop** instead of a for loop.
d) Explain how the loop works.
e) Show the output for the table of 3.

Question 16 – Kikaziki Palma

a) Input marks for **5 students in 3 subjects**.
b) Compute total and average marks using functions.
c) Sort students by average marks using a function.
d) Display the **ranked list** of students.

Question 17 – Mitala Lawrence

a) Input a number from the user.
b) Write a function isPrime(int n) to check if the number is prime.
c) Use a loop to **check numbers from 1 to n**.
d) Display **all prime numbers between 1 and n**.

Question 18 – Kityo Richard

a) Input marks for 5 students in 4 subjects.
b) Write a function grade(int mark) using if-else to assign grades: A (80–100), B (60–79), C (40–59), F (<40).
c) Display **grades for each subject for all students**.
d) Find the **student with the highest average** using a function.

Question 19 – Muwanguzi Emmanuel

a) Input **10 numbers** into an array.
b) Write a function to **reverse the array**.
c) Display the **reversed array**.
d) Explain the logic behind array reversal.

Question 20 – Wasswa Mahad

A student selects a club based on average marks.

a) Input the student's average marks.
b) Write a **switch statement**:

- 80–100 → Science Club
- 60–79 → Sports Club
- 40–59 → Drama Club
- <40 → Debate Club

c) Display the student's club choice.
d) Explain how the switch works.
e) Rewrite the program using if-else instead of switch.

Question 21 – Mukisa Luke

A school wants to summarize attendance for the term.

a) Input attendance for **15 students** (1 = present, 0 = absent) using an array.
b) Write a function totalPresent() that calculates the total number of students present.
c) Write a function totalAbsent() that calculates the total number of students absent.
d) Display **attendance percentage** for each student.

Question 22 – Mawejje Thomas

A school collects fees from students each term.

a) Input **fee paid** for 10 students.
b) Write a function paymentStatus(int fee) using if-else to determine: Full (>=500,000), Partial (100,000–499,999), None (<100,000).
c) Count and display the number of students in each category using a loop.
d) Display **each student's payment status**.

Question 23 – Kinene Malcom

a) Input a number from **5 different students**.
b) Write a function printTable(int n) to display the multiplication table for a number.
c) Use a loop in main() to call the function for all 5 numbers.
d) Rewrite the program using a **while loop** instead of a for loop.
e) Show the output for numbers 2, 5, and 7.

Question 24 – Nuwahereza Nathan

a) Input marks for **6 students in 4 subjects** using arrays.
b) Write a function totalMarks(int marks[], int n) to calculate total marks for each student.
c) Write a function averageMarks(int total) to calculate average marks.
d) Display each student's **name, total, and average**.

Question 25 – Niyonzima Allan

a) Input marks for **5 students in 5 subjects**.
b) Write a function grade(int mark) using if-else: A (80–100), B (60–79), C (40–59), F (<40).
c) Use a loop to **display grades for all students**.
d) Find the **student with the highest average** using a function.

Question 26 – Ssekitoleko Eddie

a) Input a number n from the user.
b) Write a function isPrime(int num) that returns 1 if prime, 0 if not.
c) Use a loop to **print all prime numbers from 1 to n**.
d) Display the **total number of primes** in the range.
e) Explain how your function works.

Question 27 – Nakitto Jazirah

a) Input **10 numbers** into an array.
b) Write a function reverseArray(int arr[], int n) to reverse the array.
c) Display the **reversed array**.
d) Explain the **logic used to reverse the array**.

Question 28 – Sonzi Edirisa

a) Input votes for **4 candidates** from 20 students using arrays.
b) Count the **votes for each candidate**.
c) Use if-else to **declare the winner**.
d) Display total votes and the winner.

Question 29 – Tebanyang Isaac

a) Input student average marks.
b) Write a **switch statement** to assign the student to a club:

- 80–100 → Science Club
- 60–79 → Sports Club
- 40–59 → Drama Club
- <40 → Debate Club

c) Display the **club assigned**.
d) Rewrite using **if-else statements**.
e) Explain how the switch statement works.

Question 30 – Nabulo Vicky

a) Input **fee amount** for 10 students.
b) Write a function calculateDiscount(int fee) using if-else:

- Fee > 500,000 → 10% discount
- Fee 300,000–500,000 → 5% discount
- Fee < 300,000 → 0% discount

c) Display the **discount for each student** using a loop.
d) Display the **final fee payable** after discount.

Question 31 – Akumu Mercy

A school wants to maintain **student records** including personal information and exam marks using C structures.

a) Define a **structure** called Student that contains the following fields:

- Index Number (integer)
- Name (string)
- Gender (character)
- Marks in 4 subjects (integers)

Write a C program that **declares an array of 5 students** and allows the user to input **data for each student** using a loop. (8 marks)

b) Display the **records of all students** in a formatted table showing: Index Number, Name, Gender, and marks.

c) Write a function averageMarks() that **calculates the average marks** for each student. Display the average along with the student's record.

d) Write a function highestScore() that finds and displays the student with the **highest total marks**.

e) Using **if-else statements**, assign a **grade for each student** based on their average marks:

- 80–100 → A
- 60–79 → B
- 40–59 → C
- <40 → F

Display the grade along with each student's record.

f) Rewrite the program so that **all functions take a pointer to the array of structures** instead of passing arrays directly.

g) Explain the advantages of using **structures** in this program over using separate arrays for each field.